Week 3 Progress Report

BSE4104 Emerging Trends in Software Engineering

AI-Native and Agentic Engineering Capstone Project

SME Procurement-Preparation Agent

GROUP J EVENING

1. **Work Completed Against Weekly Objectives**
    - Assembled a controlled corpus of 18 synthetic documents covering inventory policy, reorder rules, quotation validity, risk classification, human approval requirements and supporting references.
    - Created a complete Source Register with provenance, trust level and usage notes for every document.
    - Implemented document ingestion, chunking, indexing and retrieval using a vector store.
    - Built a working RAG pipeline that retrieves relevant chunks, constructs grounded context and returns answers together with Source IDs.
    - Updated the system prompt to RAG Prompt v1.1 with strict grounding and citation rules.
    - Designed and documented RAG test questions (answerable, partially answerable and deliberately unanswerable).
    - Executed the test set, recorded results, and performed peer review of the questions.
    - Documented four retrieval failures, analysed root causes and applied fixes (prompt strengthening, re-chunking, and top-k adjustment).
    - Produced a RAG architecture diagram and updated the GitHub repository with all new artefacts.

    The system can now answer policy and procedural questions from a controlled, traceable knowledge base instead of relying only on model memory.

2. Key Engineering Decisions and Rationale

DECISIONS	RATIONALE
Used only synthetic, team-created documents	Guarantees full control, avoids confidentiality issues, and makes evaluation reproducible.
Stored Source ID as metadata with every chunk	Enables accurate citation and auditability of every answer.
Forced strict “answer only from context” behaviour	Prevents hallucination and satisfies the course requirement for grounded responses.
Mixed test set	Provides balanced evaluation of retrieval quality and refusal behaviour.
Documented real failures instead of hiding them	Aligns with the course emphasis on honest evaluation and improvement.

3. Challenges Encountered and Current Response

Challenge	Response
Model sometimes cited wrong Source ID	Strengthened the prompt to allow citation only of Source IDs present in the retrieved context. Re-tested successfully.
Hallucination on partially answerable questions	Added explicit instruction to reply “Insufficient information in the knowledge base” when exact information is missing.
Relevant policy chunks ranked too low	Re-chunked key documents into smaller, focused segments and increased top-k from 4 to 6.
Weakly related chunks retrieved for unanswerable questions	Improved refusal instruction in the prompt; model now refuses more reliably.

4. Repository and Task Management Evidence

GitHub Repository: https://github.com/Nicholas-Muwonge/SME-Procurement-Agent

Key additions this week:

- knowledge/source_register.md
- knowledge/documents/ (all 18 source files)
- prompts/rag_prompt_v1.1.txt
- docs/evaluation/Week3_RAG_Evaluation_Results.md

- Updated RAG pipeline code under src/

ClickUp Board: https://app.clickup.com/1200420000004171/v/o/f/1200420000011823

All Week 3 tasks created, assigned, completed and linked to GitHub evidence.

5. **Plan for Week 4**

Focus: Tools and Function Calling

- Define at least two tools with clear purpose, input/output schemas, authorisation and failure behaviour.
- Implement tool/function calling through the application layer.
- One tool must retrieve current application data or create a low-risk simulated side effect.
- Test missing parameters, unauthorised requests, unavailable services, and unexpected tool responses.
- Add human approval before any higher-impact action.
- Deliverables: Tool Catalogue + schemas, working demonstration of tools, failure/authorisation test evidence, updated architecture diagram and Week 4 Progress Report.